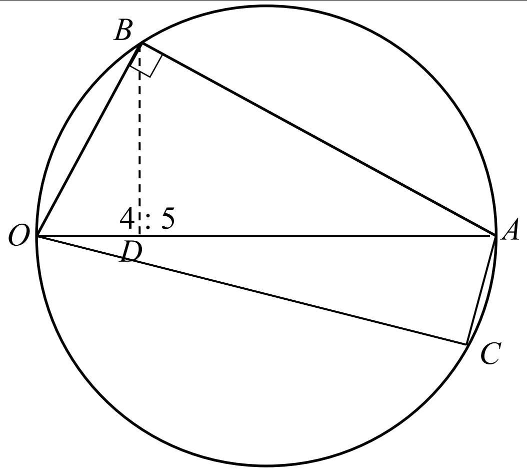
Q1	Suggested Solutions	Marking Scheme
(i)	$y=\sin^{-1}(2x)+1$ $\frac{dy}{dx}=\frac{2}{\sqrt{1-(2x)^2}}$ $=2(1-4x^2)^{-\frac{1}{2}}$ $=2\left[1+\left(-\frac{1}{2}\right)(-4x^2)+\frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{2!}(-4x^2)^2+\frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\left(-\frac{5}{2}\right)}{3!}(-4x^2)^3+\ldots\right]$ $=2\left[1+2x^2+6x^4+20x^6+\ldots\right]$ $\approx 2+4x^2+12x^4+40x^6$ The expansion of $(1-4x^2)^{-\frac{1}{2}}$ up to and including the term in x^6 is valid if $\lvert 4x^2\rvert<1$ $-\frac{1}{2}<x<\frac{1}{2}$	B1: $\frac{dy}{dx}=\frac{2}{\sqrt{1-(2x)^2}}$ M1: Apply the binomial series expansion up to x^4 B1: $\frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\left(-\frac{5}{2}\right)}{3!}$ B1: $-\frac{1}{2}<x<\frac{1}{2}$
(ii)	$y\approx\int 2+4x^2+12x^4+40x^6\ dx$ $=2x+\frac{4}{3}x^3+\frac{12}{5}x^5+\frac{40}{7}x^7+c$ When $x=0, \sin^{-1}(2x)+1=1 \Rightarrow c=1$. $\therefore \sin^{-1}(2x)\approx 1+2x+\frac{4}{3}x^3+\frac{12}{5}x^5+\frac{40}{7}x^7$	M1: Integrate each term correctly A1

<table>
<tr><th>Q2</th><th>Suggested Solutions</th><th>Marking Scheme</th></tr>
<tr><td></td><td>
</td><td></td></tr>
<tr><td></td><td>As $|\mathbf{a}|=1$, $|\mathbf{a}\cdot\mathbf{b}|$ represents either
(1) the length of projection of b onto a, or
(2) the length of OD.

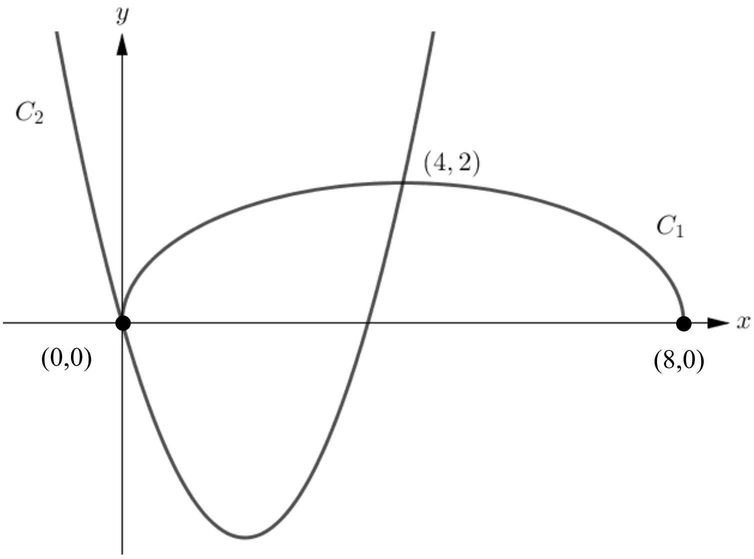
As $\overrightarrow{AB}$ and b are perpendicular (properties of semi-circle)
$\mathbf{b}\cdot(\mathbf{b}-\mathbf{a})=0$
$|\mathbf{b}|^2-\mathbf{a}\cdot\mathbf{b}=0$
$\mathbf{a}\cdot\mathbf{b}=|\mathbf{b}|^2$</td><td>B1

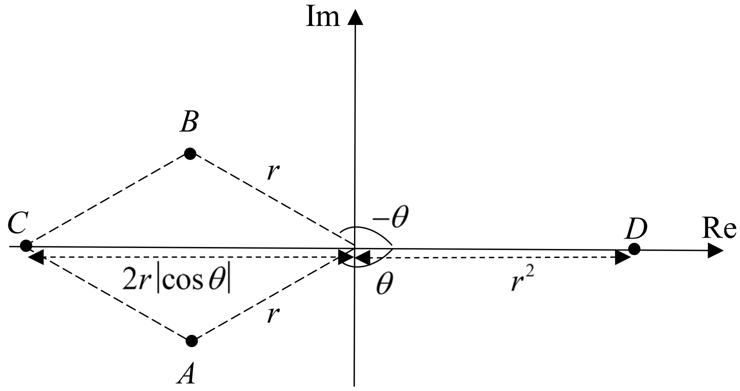
B1: perpendicular implies dot product = 0

B1: (a.g.)</td></tr>
<tr><td></td><td>OD : DA = 4 : 5 and $\left|\overrightarrow{OA}\right|=|\mathbf{a}|=1$
Let angle AOB = θ.
$\cos\theta=\frac{\mathbf{a}\cdot\mathbf{b}}{|\mathbf{a}||\mathbf{b}|}=\frac{|\mathbf{b}|^2}{|\mathbf{a}||\mathbf{b}|}=|\mathbf{b}|$.
Considering triangle AOB, $\cos\theta=\frac{|\mathbf{b}|}{|\mathbf{a}|}$.
$\frac{|\mathbf{b}|}{|\mathbf{a}|}=\frac{\frac{4}{9}|\mathbf{a}|}{|\mathbf{b}|}\Rightarrow|\mathbf{b}|^2=\frac{4}{9}|\mathbf{a}|^2=\frac{4}{9}$ as a is a unit vector
Hence $\cos\theta=\frac{2}{3}$.</td><td>M1: use of scalar product formula and part (i)

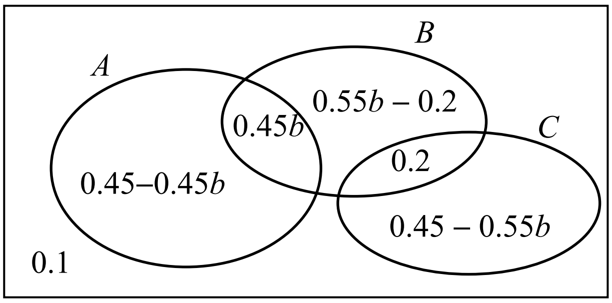
A1</td></tr>
<tr><td></td><td>Area of triangle OCD = $\frac{1}{2}(OD)$ (height)
= $\frac{1}{2}\left(\frac{4}{9}\right)(0.4)=\frac{8}{90}$
OR $=\frac{1}{2}\left|\overrightarrow{OC}\times\overrightarrow{OD}\right|=\frac{1}{2}\left|\mathbf{c}\times\frac{4}{9}\mathbf{a}\right|=\frac{1}{2}\left(\frac{4}{9}\right)(0.4)=\frac{8}{90}=\frac{4}{45}$</td><td>M1: Area of triangle or relate to ratio
A1: $\frac{4}{45}$</td></tr>
</table>

Q3	Suggested Solutions	Marking Scheme
(i)	$a_2 = 3.5a_1 - 1.2$ $a_3 = 3.5a_2 - 1.2$ $= 3.5(3.5a_1 - 1.2) - 1.2$ $= 3.5^2 a_1 - 1.2(3.5) - 1.2$ $= 3.5^2 a_1 - 1.2(1 + 3.5)$ $a_4 = 3.5\left[3.5^2 a_1 - 1.2(1+3.5)\right] - 1.2$ $= 3.5^3 a_1 - 1.2(1 + 3.5 + 3.5^2)$ $a_n = 3.5^{n-1} a_1 - 1.2(1 + 3.5 + 3.5^2 + \ldots + 3.5^{n-2})$ $= 3.5^{n-1} a_1 - 1.2\left(\frac{(1)(3.5^{n-1} - 1)}{3.5 - 1}\right)$ $= 3.5^{n-1} a_1 - 0.48(3.5^{n-1} - 1)$ (Shown)	B1: write out a_2 and a_3 correctly M1: recognising GP pattern A1: use of GP sum formula and have clear working leading to answer (a.g.)
(ii)	By the end of week 3, the number of cells have increased by 250% in from the beginning of week 3. No. of particles in <u>millions</u> before the cells are transferred to another laboratory $= a_4 + 1.2$ $= 3.5^{4-1}(0.5) - 0.48(3.5^{4-1} - 1) + 1.2$ $= 2.5375$	M1: Find a_4 A1: 2.5375 millions or 2,537,500
(iii)	Let the number of cells (in millions) in Lab *B* at the start of the *n*th week is given by d_n $d_1 = 0$ $d_2 = 0.5$ $d_3 = 0.5 + 5$ $d_4 = 0.5 + 5 + (5 + 5)$ $= 0.5 + 5 + 2(5)$ $d_n = 0.5 + 5 + 2(5) + 3(5) + \ldots + (n-2)(5)$ $= 0.5 + \frac{n-2}{2}\left(10 + (n-2-1)(5)\right)$ $= 0.5 + \frac{n-2}{2}\left(10 + 5(n-3)\right)$ $3.5^{n-1}(0.5) - 0.48(3.5^{n-1} - 1) + 0.5 + \frac{n-2}{2}\left(10 + 5(n-3)\right) \geq 60$ By G.C., least $n = 6$ $\therefore$ The earliest week is week 6.	M1: Apply AP formula with $(n-2)$ no. of terms M1: Add GP and AP and equate to 60 A1: Week 6

Q4	Suggested Solutions	Marking Scheme
		B1: Shape of C_1 for $-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$ B1: $(0,0)$, (8, 0)
(a)	Area bounded by C_1 and C_2 $= \int_0^4 y_1 - y_2 \, dx$ $= \int_{-\frac{\pi}{2}}^{0} 2\cos\theta(4\cos\theta)\,d\theta - \int_0^4 x^2 - \frac{7}{2}x \, dx$ $= 4\int_{-\frac{\pi}{2}}^{0} 2\cos^2\theta \, d\theta - \left[\frac{x^3}{3} - \frac{7x^2}{4}\right]_0^4$ $= 4\int_{-\frac{\pi}{2}}^{0} (1+\cos 2\theta)\,d\theta - \left[\frac{x^3}{3} - \frac{7x^2}{4}\right]_0^4$ $= 4\left[\theta + \frac{1}{2}\sin 2\theta\right]_{-\frac{\pi}{2}}^{0} - \left[\frac{64}{3} - \frac{7\times 16}{4}\right]$ $= \left(2\pi + \frac{20}{3}\right)$ units^2	B1: Shape of C_2 (found in diagram above) with (4, 2) intersecting at the max point of C_1. M1: Correct set up with limits of the area for integration (difference of areas) M1: parametric set up (as per substitution method) M1: apply trigo identity to integrate of $\cos^2\theta$ A1: Ans (Correct application of the formula for area of ellipse will result in loss of second and third M1.)
(b)	Since C_2 is a quadratic curve with x-intercepts 0 and a, in order for C_1 and C_2 to meet at two distinct points, $0 < a \le 8$.	B1. Ans. Also accept $a \le 8$

Q5	Suggested Solutions	Marking Scheme
(i)	$\lvert w^* z\rvert = r^2 \Rightarrow \lvert w^*\rvert\lvert z\rvert = r^2$ $\Rightarrow \lvert w\rvert\lvert z\rvert = r^2$ $\Rightarrow \lvert w\rvert = r$ $\arg\left(-\dfrac{w}{z^3}\right) = \pi - 4\theta \Rightarrow \arg(-w) - \arg(z^3) = \pi - 4\theta$ $\Rightarrow \arg(-1) + \arg(w) - 3\arg(z) = \pi - 4\theta$ $\Rightarrow \pi + \arg(w) - 3\theta = \pi - 4\theta$ $\Rightarrow \arg(w) = -\theta$	M1: Correctly applying the properties of mod A1: $\lvert w\rvert = r$ B1: apply $\arg(-w) = \pi + \arg(w)$ B1: apply $\arg(a^n) = n\arg(a)$ and $\arg\left(\dfrac{a}{b}\right) = \arg(a) - \arg(b)$
(ii)	 Note that since z and w are conjugates of each other, $z + w = 2\,\mathrm{Re}(z) = 2r\cos\theta$ and $zw = \lvert z\rvert^2 = r^2$ must therefore lie on the real axis. $\lvert z+w\rvert = -2r\cos\theta$ or $2r\lvert\cos\theta\rvert$.	B1: For correctly positioned point *A* or *B*, with labelling of mod and arg. B1: for points *A* and *B* being reflection of each other in the real axis B1: For point *C*, it should be part of a rhombus *OACB* and lie on real axis. B1: For point *D* (must be on positive real axis) with labelling of mod r^2, *OD* further from origin than *OA* and *OB*. B1 Allow up to 1 slip for labelling.
(iii)	Area of the quadrilateral *ACBD* $= 2\times\dfrac{1}{2}(-r\sin\theta)(-2r\cos\theta + r^2)$ $= -r\sin\theta(-2r\cos\theta + r^2)$ $= 2r^2\sin\theta\cos\theta - r^3\sin\theta$ $= r^2\sin 2\theta - r^3\sin\theta$	B1: $AB = -2r\sin\theta$ B1: $2\times\dfrac{1}{2}\left(\dfrac{1}{2}AB\right)(-2r\cos\theta + r^2)$ or $\dfrac{1}{2}(AB)(-2r\cos\theta + r^2)$ leading to the given result

Q6	Suggested Solutions	Marking Scheme
(i)	$\bar{x}=\dfrac{1261.05}{105}=12.01$ $s^2=\dfrac{1}{n-1}\left[\sum x^2-\dfrac{\left(\sum x\right)^2}{n}\right]$ $=\dfrac{1}{104}\left[15145.47-\dfrac{(1261.05)^2}{105}\right]$ $=0.0024951923$ $=0.00250$ (3 s.f.)	B1: $\bar{x}=12.01$ B1: $\dfrac{519}{208000}$
(ii)	$\mathrm{H}_0:\mu=12$ versus $\mathrm{H}_1:\mu>12$ Under H_0, $\bar{X}\sim \mathrm{N}\left(12,\ \dfrac{s^2}{105}\right)$ approximately by Central Limit Theorem since n = 105 is large. Upper-tailed test at 3% level of significance. Critical region: Reject H_0 if p-value < 0.03 p-value = 0.0201157096 < 0.03 Reject H_0 and conclude that there is sufficient evidence at 3% level of significance that the mean diameter, after the modification, of the spherical pellets has increased.	B1: alternative hypothesis and $\bar{X}$ distribution (must state approximately and/or Central Limit Theorem) B1: p-value = 0.0201 < 0.03 or test stat. 2.05 > 1.88 A1: Correct and complete conclusion
(iii)	There is a probability of 0.0201 that the sample mean diameter is more than 12.01 mm, given that the population mean diameter is 12 mm.	B1: can neglect numerical probability value of 0.0201
(iv)	Since the sample size is 105 and it is considered as large, hence we can apply Central Limit Theorem to approximate the distribution of $\bar{X}$ to be normal.	B1: need to state that sample size is large / >20 and Central Limit Theorem (spelt in full) and $\bar{X}$ normal

Q7	Suggested Solutions	Marking Scheme
(i)	$\mathrm{P}(A' \cap B) = \mathrm{P}(B) - \mathrm{P}(A \cap B)$ $= b - 0.45b$ $= 0.55b$ $\mathrm{P}(A') \times \mathrm{P}(B) = (1-0.45) \times b$ $= 0.55b$ As $\mathrm{P}(A' \cap B) = \mathrm{P}(A') \times \mathrm{P}(B)$, A' and B are independent events.	Must not assume A' and B are independent events M1: Obtain values for $\mathrm{P}(A' \cap B)$ and $\mathrm{P}(A') \times \mathrm{P}(B)$ A1: Show that $\mathrm{P}(A' \cap B) = \mathrm{P}(A') \times \mathrm{P}(B)$, and concluding A' and B are independent events.
(ii)	$\mathrm{P}(A \cap B') = 0.45 - 0.45b$ $\mathrm{P}(A' \cap B \cap C') = 0.55b - 0.2$	B1 B1
(iii)	$\mathrm{P}(B' \cap C) = (1-0.1) - (0.45 - 0.45b) - b$ $= 0.45 - 0.55b$  $0.55b - 0.2 \geq 0$ and $0.45 - 0.45b \geq 0$ and $0.45 - 0.55b \geq 0$ $\Rightarrow b \geq \frac{4}{11}$ and $b \leq 1$ and $b \leq \frac{9}{11}$ $\Rightarrow \frac{4}{11} \leq b \leq \frac{9}{11}$ $\Rightarrow \frac{9}{55} \leq 0.45b \leq \frac{81}{220}$ Maximum possible value of $\mathrm{P}(A \cap B)$ is $\frac{81}{220}$. Minimum possible value of $\mathrm{P}(A \cap B)$ is $\frac{9}{55}$.	M1: Calculate $\mathrm{P}(B' \cap C)$ in terms of b M1: Setup inequalities to obtain the range of values for b A1: Greatest is $\frac{81}{220}$ A1: Least is $\frac{9}{55}$

Q8	Suggested Solutions	Marking Scheme
(a)	$\binom{4}{1}\times\binom{3}{1}\times\binom{2}{1}\times\binom{1}{1}=24$ $\binom{4}{1}$: select 1 shape of colour 1; $\binom{3}{1}$: select 1 remaining shape of colour 2; $\binom{2}{1}$: select 1 remaining shape of colour 3; $\binom{1}{1}$: select 1 remaining shape of colour 4 No. of arrangements of 4 chosen blocks $=4!$ $\therefore$ No. of arrangements $=24\times 4!=576$.	B1: 24 or 4! Or 4 x 3 x 2 x 1 B1
(b)	<u>Case 1: 2 pentagons of different colours or same colour, 1 triangle and 1 parallelogram, 1 square</u> $\left[\binom{4}{2}+\binom{4}{1}\right]\times\binom{4}{1}\times\binom{4}{1}\times\binom{4}{1}=640$ $\left[\binom{4}{2}+\binom{4}{1}\right]$: Two cases: select 2 pentagons of different colours and of the same colour; $\binom{4}{1}$: select 1 triangle; $\binom{4}{1}$: select 1 parallelogram; $\binom{4}{1}$: select 1 square <u>Case 2: 2 pentagons of different colours or same colour, 2 triangles and 1 parallelogram</u> $\left[\binom{4}{2}+\binom{4}{1}\right]\times\binom{4}{2}\times\binom{4}{1}=240$ $\left[\binom{4}{2}+\binom{4}{1}\right]$: Two cases: select 2 pentagons of different colours and of the same colour; $\binom{4}{2}$: select 2 triangles of different colours; $\binom{4}{1}$: select 1 parallelogram <u>Case 3: 2 pentagons of different colours or same colour, 1 triangle and 2 parallelograms</u> $\left[\binom{4}{2}+\binom{4}{1}\right]\times\binom{4}{1}\times\binom{4}{2}=240$ $\left[\binom{4}{2}+\binom{4}{1}\right]$: Two cases: select 2 pentagons of different colours and of the same colour; $\binom{4}{1}$: select 1 triangle; $\binom{4}{2}$: select 2 parallelograms of different colours Total number of ways $=640+240+240$ $=1120$	M1: Consider cases (take all pattern blocks except pentagons distinguishable) M1: Consider whether the 2 pentagons are of same colour or 2 different colours i.e. $\left[\binom{4}{2}+\binom{4}{1}\right]$ A1: 1120
Last part	Step 1: Number of ways to arrange the pentagons (as one unit), parallelograms and squares in a circle $=\frac{9!}{9}$ Step 2: Number of ways to arrange all pentagons next to each other $=\frac{8!}{2!2!2!2!}$ Step 3: Number of ways to slot in the distinct triangles $=\binom{9}{4}\times 4!$ Total number of arrangements	B1: $\frac{9!}{9}$ or $(9-1)!$ M1: account for repetition for 4 pairs of pentagons in denominator M1: account for slotting 4 triangles into 9 objects (either row or circle)

Q8	Suggested Solutions	Marking Scheme
	$=\dfrac{9!}{9}\times\dfrac{8!}{2!2!2!2!}\times\begin{pmatrix}9\\4\end{pmatrix}\times 4!$ $=3.072577536\times10^{11}$ $=3.07\times10^{11}$	A1: 3.07×10^{11} or 307257753600

<table>
<tr><th>Q9</th><th>Suggested Solutions</th><th>Marking Scheme</th></tr>
<tr><td>(i)</td><td>

$\mathrm{P}(X=2)=\frac{1}{6}\times\frac{1}{6}\times\frac{5}{6}\times 3=\frac{15}{216}=\frac{5}{72}$

$\mathrm{P}(X=0)=\frac{5}{6}\times\frac{5}{6}\times\frac{5}{6}=\frac{125}{216}$

$\mathrm{P}(X=1)=\frac{1}{6}\times\frac{5}{6}\times\frac{5}{6}\times 3=\frac{75}{216}=\frac{25}{72}$

$\mathrm{P}(X=3)=\frac{1}{6}\times\frac{1}{6}\times\frac{1}{6}=\frac{1}{216}$

X	0	1	2	3
$\mathrm{P}(X=x)$	$\frac{125}{216}$	$\frac{25}{72}$	$\frac{5}{72}$	$\frac{1}{216}$

</td><td>

B1: $\frac{1}{6}\times\frac{1}{6}\times\frac{5}{6}\times 3$

B1: Any 2 others correct

B1: All correct and answers on table. (Answers should be in fractions but need not be in lowest terms. Accept $\frac{75}{216},\frac{15}{216}$)

</td></tr>
<tr><td>(ii)</td><td>

$\mathrm{E}(X)=0\left(\frac{125}{216}\right)+1\left(\frac{25}{72}\right)+2\left(\frac{5}{72}\right)+3\left(\frac{1}{216}\right)$

$=\frac{1}{2}$

$\mathrm{Var}(A)=\mathrm{E}(A^2)-\left(-\frac{101}{216}\right)^2$

$=0^2\left(\frac{125}{216}\right)+1^2\left(\frac{25}{72}\right)+2^2\left(\frac{5}{72}\right)+3^2\left(\frac{1}{216}\right)-\left(\frac{1}{2}\right)^2$

$=\frac{5}{12}$ or 0.417

Let $\overline{A}=\frac{A_1+\ldots+A_{30}}{30}$

Since the distribution of A is not a normal distribution and $n=30$ is large enough, by Central Limit Theorem,

$\overline{A}\sim \mathrm{N}\left(\frac{1}{2},\, {}^{\frac{5}{12}}\!\big/_{30}\right)$ approximately.

$\mathrm{P}(\overline{A}>1)=0.00001105257393$

$=0.0000111$ (3 s.f.)

</td><td>

B1: $\frac{1}{2}$

M1: Find $\mathrm{Var}(A)$ with attempt to calculate $\mathrm{E}(A^2)$

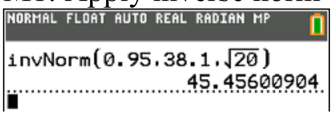
B1: Correct justification for approximating $\overline{A}$ to a normal distribution.

A1: 0.0000111 or 1.11×10^{-5}

</td></tr>
</table>

Q9	Suggested Solutions	Marking Scheme
(iii)	Let A be the net losses in dollars for each game <table><tr><td>a</td><td>$-k$</td><td>0</td><td>k</td><td>$9k$</td></tr><tr><td>$\mathrm{P}(A=a)$</td><td>$\frac{125}{216}$</td><td>$\frac{25}{72}$</td><td>$\frac{5}{72}$</td><td>$\frac{1}{216}$</td></tr></table> $\mathrm{P}(A_1+A_2+A_3<0)$ $=\mathrm{P}(\text{all "}-k\text{"})+\mathrm{P}(2\text{"}-k\text{","}0\text{"})+\mathrm{P}(2\text{"}-k\text{","}k\text{"})+\mathrm{P}(\text{"}-k\text{"},2\text{"}0\text{"})$ $=\left(\frac{125}{216}\right)^3+\frac{3!}{2!}\left(\frac{125}{216}\right)^2\left(\frac{25}{72}\right)+\frac{3!}{2!}\left(\frac{125}{216}\right)^2\left(\frac{5}{72}\right)+\frac{3!}{2!}\left(\frac{125}{216}\right)\left(\frac{25}{72}\right)^2$ $=0.821740406$ $=0.822$ (3 s.f.)	M1: Consider any one of the appropriate cases which needs to multiply by $\frac{3!}{2!}$ M1: consider at least 3 correct cases A1 Accept answer if the values of a include $10k$

Q10	Suggested Solutions	Marking Scheme
(a)(i)	$6p(1-p)=0.8466$ $p^2-p-0.1411=0$ $\therefore p=0.17$ (since $p<0.5$)	B1: $p=0.17$
(a)(ii)	$\mathrm{P}(X\neq 6)=\frac{117648}{117649}$ $1-\mathrm{P}(X=6)=\frac{117648}{117649}$ $\mathrm{P}(X=6)=\frac{1}{117649}$ $\binom{6}{6}p^6(1-p)^0=\frac{1}{117649}$ $p^6=\frac{1}{117649}$ $\therefore p=\frac{1}{7}$	M1: Express $\mathrm{P}(X=6)$ in algebraic form A1: $p=\frac{1}{7}$
(b)(i)	The event that a customer is female is independent of the event that another customer is female. The probability that any one customer is female is the same throughout for the first n customers.	B1 B1
(b)(ii)	Let Y be the number of female customers out of 12. $Y\sim \mathrm{B}(12,0.65)$ Required probability $=\mathrm{P}(Y=7)\times 0.65\times(1-0.65)^2$ $=0.2039195668\times 0.65\times 0.35^2$ $=0.0162370955$ $=0.0162$ (3 s.f.)	P(Var): If random variable is not clearly defined. M1: Calculate probability that <u>7 out of 12 customers</u> are females B1: $(1-0.65)^2$ A1: 0.0162
(b)(iii)	Let F be the number of female customers out of 15. $F\sim \mathrm{B}(15,0.65)$ $\mathrm{P}(F\geq 8)=1-\mathrm{P}(F\leq 7)$ $=1-0.1132311308$ $=0.8867688692$ Let W be the number of days out of 7 with at least 8 female customers who visits the salon within the 1st hour. $W\sim \mathrm{B}(7,\mathrm{P}(F\geq 8))$ $\mathrm{P}(W=3)=0.0040120168$ $=0.00401$ (3 s.f.)	P(Var): If random variable is not clearly defined. M1: Apply $\mathrm{P}(F\geq 8)=1-\mathrm{P}(F\leq 7)$ M1: Identify new random variable in terms of number of days A1: 0.00401

Q12	Suggested Solutions	Marking Scheme
(a)	Let X be the random variable denoting the BMI for a 18 year old boy in Singapore. $X \sim \mathrm{N}\left(\mu, \sigma^2\right)$ $\mathrm{P}\left(X < 16.7\right) = 0.05$ $\mathrm{P}\left(Z < \frac{16.7-\mu}{\sigma}\right) = 0.05$ $\frac{16.7-\mu}{\sigma} = -1.644853626$ $\mu - 1.6449\sigma = 16.7 --- (1)$ $\mathrm{P}\left(X > 27.4\right) = 0.1$ $\mathrm{P}\left(Z > \frac{27.4-\mu}{\sigma}\right) = 0.1$ $\frac{27.4-\mu}{\sigma} = 1.281551567$ $\mu + 1.2816\sigma = 27.4 --- (2)$ Solving (1) and (2) simultaneously, we have $\mu = 22.71418212$ and $\sigma = 3.656363113$ $= 22.7$ (3 s.f.) $= 3.66$ (3 s.f.)	M1: Form at least 1 correct equation M1: standardisation A1: $\mu = 22.7$ A1: $\sigma = 3.66$
(a)(ii)	No. based on the data, the group will have a distribution that is bimodal and thus cannot be normal. Alternative : It cannot be assumed that the BMI for 13 year old boys in Singapore is normally distributed. Hence no conclusion can be made about the combined sample.	B1
(b)(i)	Let X be the time taken for food preparations. Let Y be the time taken for the delivery time by a delivery man on bicycle. Let t be the time taken for food prep such that there is a 95% chance that the food is delivered by 7pm. $X \sim N\left(20, 4^2\right)$ $Y \sim N\left(18.1, 2^2\right)$ $X + Y \sim N\left(38.1, 2^2 + 4^2\right)$ $\mathrm{P}(X + Y < t) \geq 0.95$ $t \geq 45.5$ mins She should place her order latest by 6.14pm.	M1: Correct method to find mean and variance A1: Both mean and variance correct. M1: Apply inverse norm 

Q12	Suggested Solutions	Marking Scheme
		A1: Correct answer P: Minus 1 if rv are no defined correctly.
(b)(ii)	Let W be the time taken for the delivery time by a deliveryman on motorbike. $W \sim N\left(15, 4^2\right)$ $Y \sim N\left(18.1, 2^2\right)$ $D = 1.2\left(W_1 + \ldots W_{10}\right) - 1\left(Y_1 + \ldots Y_{10}\right) \sim \mathrm{N}\left(-1, 270.4\right)$ $\mathrm{P}(D > 0) = 0.476$ Since there is a higher probability that a delivery man on bicycle earns more than one on motorbike. The restaurant owner should employ those on motorbike to save more money.	M1: Correct method to find mean and variance. A1: Both mean and variance correct. B1: Find an appropriate probability. B1: Correct conclusion. P: Minus 1 if the rv is not properly defined.